

AI StudyBuddy

AI-Powered Study Management and Learning Support System

PHASE 3

Project Design

Role	Team Member
Team Leader	Kishore T
Team Member	Logeshwaran D
Team Member	Manikandan M
Team Member	Mohamed Safir M

Submitted as a phase-wise project submission.

1. System Architecture

The backend follows a modular REST architecture. Client requests are handled by Express.js; authentication is protected through JWT middleware; application logic is separated into routes and controllers; MongoDB stores application data; and the AI service communicates with Google Gemini.

2. Main Components

- Client/API testing layer
- Express server
- JWT authentication middleware
- Route and controller modules
- MongoDB/Mongoose database layer
- AI service layer for Gemini-powered learning features
- Administrator user-management operations

3. Project Structure Evidence

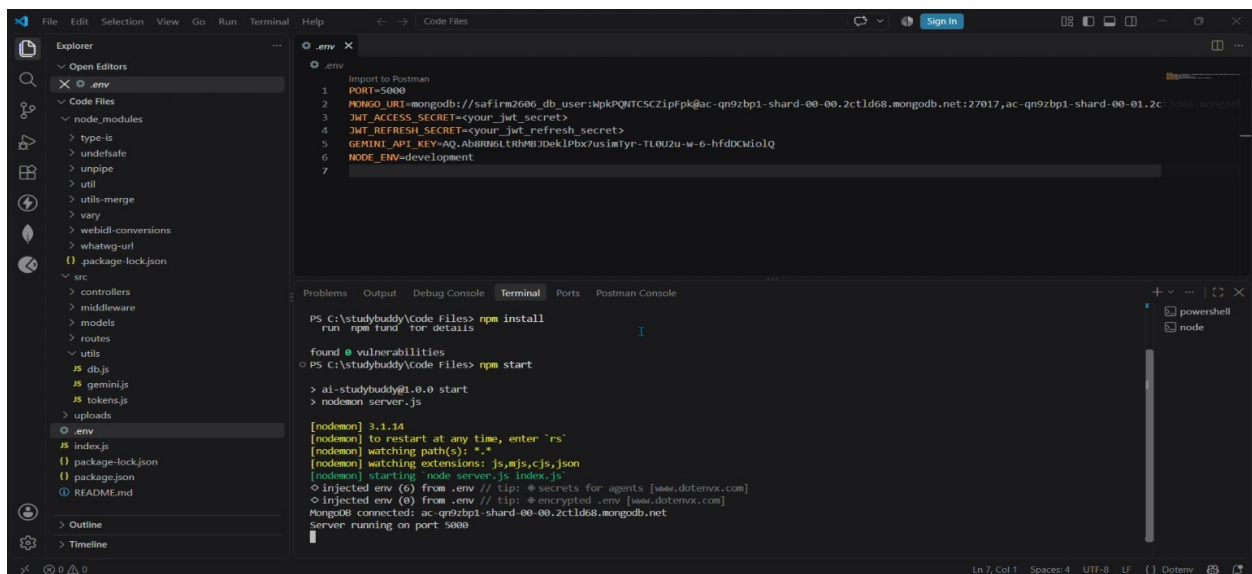


Figure 3.1: Visual Studio Code project structure, environment configuration and successful server startup.

4. User Flow

Register/Login → JWT authentication → Upload study material → Select AI feature → AI processing → Display generated learning resource.

5. Data Entities

- User: identity, credentials and role.
- Study Material: uploaded material linked to a user.
- Summary: generated summary linked to study material.
- Flashcard: generated question/answer items.
- Quiz: generated questions and answers.

- Study Plan: personalized plan linked to a user.